

Full-production P3: quantitative smoothing and dealiased evolution

TECT verification-first repository · claim A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM

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1. Purpose and scope

This supplement resolves the two previously qualitative P3 inputs: an explicit positive-time H^6 solution-ball majorant and an explicit finite-time error bound for the dealiased exact Galerkin flow. The scope is the canonical P1/P2 three-component gradient flow on the fixed 16^3 torus, positive regularisers, $\eta_{\text{shell}} = 0$, initial H^2 radii $R = 0.5, 1, 2$, and $0.02 = \tau \leq t \leq T = 0.2$.

For merely H^2 initial data the Galerkin comparison is restarted at positive time: $u_N(\tau) = P_N u(\tau)$. This is essential. No algebraic rate from $t = 0$, finite-oversampling exactness for the rational Class-II residual, or historical Sector-B solver certificate is asserted.

1 Explicit Fourier constants

Use

$$\|u\|_{H^{2m}}^2 = |\mathbb{T}^3| \sum_k (1 + |k|^4)^m |\widehat{u}_k|^2.$$

The max-shell count $24m^2 + 2$ and an integral tail give

$$\|u\|_{\infty} \leq E_2 \|u\|_{H^2}, \quad E_2 = 0.329926816704,$$

and Hausdorff–Young with the independently enclosed series $\sum k_i^4 / (1 + |k|^4)^2$ gives

$$\|\partial_i u\|_{L^4} \leq E_{14} \|u\|_{H^2}, \quad E_{14} = 0.533391254984.$$

The Fourier weight inequality $w(k) \leq 4\{w(\ell) + w(k - \ell)\}$, followed by Young convolution, yields the deliberately enlarged algebra constant

$$\|fg\|_{H^2} \leq A_2 \|f\|_{H^2} \|g\|_{H^2}, \quad A_2 = 8E_2 = 2.63941453363. \quad (2.1)$$

The P2 lower graph constant is $c_L = 0.2048572626782363$. A direct upper bound on the scalar symbol plus the family/lock norm gives $C_L = 2.1566713536$, hence

$$c_L \|u\|_{H^2}^2 \leq \langle Lu, u \rangle \leq C_L \|u\|_{H^2}^2. \quad (2.2)$$

2 Class-II coefficient envelopes

Write $q_A = m_A / (\rho + \varepsilon_\rho)$, with $\varepsilon_\rho = 10^{-12}$, $J_A = A_J(u) \nabla u$, and $K_A = A_K(u) \nabla u$. The embedded Pauli generators have norm one, so $\|A_J\| \leq 2|u|$ and $\|A_K\| \leq 4|u|$.

For $|u| \leq A$, derivatives of $(\rho + \varepsilon_\rho)^{-1}$ through order $j \leq 3$ are bounded by

$$I_j = j!(2A + 2)^j \varepsilon_\rho^{-(j+1)}. \quad (3.1)$$

Leibniz convolution with the quadratic moment gives explicit Q_j for $D^j q_A$. A second Leibniz convolution gives derivative envelopes G_j for the Class-II metric

$$B(u) = \sum_A \{a A_J^* A_J + b(A_J^* A_K + A_K^* A_J) + c A_K^* A_K\}.$$

All component, spatial, and generator contractions are then enlarged by the declared factor 108. This overcount preserves signs and makes every dependence on the small density floor visible instead of hiding it in a Moser symbol.

Consequently the full lower-order map has computable bounds

$$\|N(u)\|_{L^2} \leq K_0(M_2), \quad \|N(u) - N(v)\|_{L^2} \leq L_0(M_2)\|u - v\|_{H^2}, \quad (3.2)$$

and, after an H^4 bound is available,

$$\|N(u)\|_{H^2} \leq K_2(M_2, M_4), \quad \|N(u) - N(v)\|_{H^2} \leq L_2(M_2, M_4)\|u - v\|_{H^4}. \quad (3.3)$$

The executable records every intermediate Q_j, G_j, K_s, L_s .

3 Two quantitative endpoint cancellations

For a positive self-adjoint L ,

$$\|L^\alpha e^{-tL}\| \leq (\alpha/(et))^\alpha. \quad (4.1)$$

Choose $\alpha = 3/4$, $\theta = \alpha - 1/2 = 1/4$, and $\delta = \tau/4$. The mild formula first gives an explicit $X_{3/4}$ bound, where $X_s = D(L^s)$. It also makes u Holder continuous in $X_{1/2}$ and, using L_0 , makes $N(u)$ Holder continuous in L^2 .

The endpoint identity

$$L \int_a^t S(t-s)f(s)ds = (I - S(t-a))f(t) + \int_a^t LS(t-s)\{f(s) - f(t)\}ds \quad (4.2)$$

changes the nonintegrable r^{-1} kernel into $r^{\theta-1}$. Equations (2.2)–(4.2) give an explicit M_4 on $[\tau/2, T]$.

Repeat the same construction on the base space H^2 , using (3.3). The second endpoint cancellation produces $X_{3/2}$ and therefore

$$\sup_{\|u_0\|_{H^2} \leq R} \sup_{\tau \leq t \leq T} \|u(t)\|_{H^6} \leq B_6(R, \tau, T). \quad (4.3)$$

The bounds are enormous because (3.1) treats the 10^{-12} floor in the worst possible direction, but they are finite and fully computable. For $R = 0.5, 1, 2$, respectively,

$$\log_{10} B_6 \leq 515.378, \quad 515.751, \quad 517.320. \quad (4.4)$$

4 Residual and aliasing constant

The omitted Fourier cube obeys

$$\|(I - P_N)u\|_{H^2} \leq (L_{\max}/\pi)^4 N^{-4} \|u\|_{H^6}. \quad (5.1)$$

For periodic sampling, Cauchy–Schwarz over aliases and

$$\sum_{m \geq 1} \frac{24m^2 + 2}{m^4} = 4\pi^2 + \frac{\pi^4}{45}$$

give

$$\|P_N(I_N - I)f\|_{L^2} \leq A_{\text{alias}}N^{-2}\|f\|_{H^2}, \quad A_{\text{alias}} = 167.383081344. \quad (5.2)$$

Combining (3.2)–(5.2) yields the now evaluated statement

$$\|P_N R(u) - R_N^C(P_N u)\|_{L^2} \leq C(R, \tau, T)N^{-2}. \quad (5.3)$$

For $R = 0.5, 1, 2$, the conservative logarithmic bounds are

$$\log_{10} C \leq 576.355, \quad 576.770, \quad 578.513. \quad (5.4)$$

5 Dealiased finite-time evolution

Let u_N be the exact Galerkin flow

$$\partial_t u_N + Lu_N + P_N N(u_N) = 0, \quad qquad u_N(\tau) = P_N u(\tau). \quad (6.1)$$

This is the mathematically dealiased object. For the rational Class-II term, a fixed finite oversampling factor is not called exact Galerkin.

The restricted gradient flow has the same energy identity and H^2 envelope. For $e = u_N - P_N u$, coercivity and (3.2) give

$$\frac{d}{dt}\|e\|_{L^2} \leq a_R\|e\|_{L^2} + L_0(M_2)(L_{\max}/\pi)^4 B_6 N^{-4}, \quad a_R = \frac{L_0(M_2)^2}{2c_L}. \quad (6.2)$$

Since $e(\tau) = 0$, Gronwall yields

$$\sup_{\tau \leq t \leq T} \|u_N(t) - P_N u(t)\|_{L^2} \leq E(R, \tau, T)N^{-4}, \quad (6.3)$$

where

$$E = L_0(L_{\max}/\pi)^4 B_6 \frac{e^{a_R(T-\tau)} - 1}{a_R}. \quad (6.4)$$

The implementation stores $\log_{10} E$ with Decimal arithmetic because the worst-case exponential is far beyond floating-point range. This proves finite-time convergence but is not a practical error budget at present grids.

6 Executable and independent audits

The primary audit passes 15/15. The independent NumPy audit does not import the majorant implementation; it reconstructs the linear spectrum, samples the universal semigroup inequality, checks the N^{-4} projection tail and alias shell identity, probes exact Class-II q directional derivatives against the analytic envelopes, and recomputes the Decimal logarithms. It passes 10/10. The integrated P3 verifier consequently passes 104/104.

7 Devil’s-advocate review

1. **A finite constant is automatically a useful numerical error bar. UPHELD as false.** The floor-based bounds are astronomically large; they prove convergence but do not certify practical N32/N64/N128 runs.
2. **The H^6 estimate is valid at $t = 0$ for H^2 data. UPHELD as false.** Both (4.3) and the Galerkin comparison require $\tau > 0$; (6.1) is explicitly restarted at τ .

3. **Three-halves dealiasing is exact for the Class-II backend. UPHELD as false.** The rational density denominator has no finite polynomial bandwidth. Only $P_N N(u_N)$ is called exact Galerkin.
4. **The 10^{-12} floor may be ignored because $|q_A| \leq 1$. VALID WITH MITIGATION.** The value is bounded, but its derivatives enter Moser/Lipschitz constants; (3.1) retains the full floor dependence.
5. **Large hardcoded numbers mask the derivation. DISMISSED.** Production values are read from pinned manifests; all reported majorants are recomputed and labelled DERIVED.
6. **This result alone authorises T6. UPHELD as a governance error.** A final self-contained theorem package, PUBLISHED reproduction bundle, and separate tier review are still required.

8 What remains open

The two selected mathematical blockers are closed in the stated positive-time exact-Galerkin scope. Remaining promotion work is packaging and tier review. Separately, sharp constants, a rate from $t = 0$ for merely H^2 data, and a certified finite-oversampling implementation for the rational Class-II term remain research improvements, not hidden premises of (4.3)–(6.3).

Result footer

Result ID:	A3-FULL-PRODUCTION-DISCRETIZATION-CONTINUUM
Precise statement:	Explicit B6 and residual C on $[\tau, T]$, plus N^{-4} convergence of exact Galerkin restarted at τ .
Scope:	Fixed T3, canonical P1/P2, $\eta_{\text{shell}}=0$, $R=0.5/1/2$, $\tau=0.02$, $T=0.2$, exact P_N nonlinear projection.
Dependencies:	A1 production functional; A2 T6 well-posedness; P3 energy-to- H^2 envelope.
Evidence grade:	ANALYTIC + EXECUTED + CONDITIONAL.
Reproduction command:	python codes/foundations/a3_full_production_quantitative_majorant.py
Expected output:	15/15 PASS; A3-FULL-QUANTITATIVE-MAJORANT-PASS.
Falsification gate:	Failed coefficient envelope, endpoint constant, Fourier tail, alias sum, or independent audit.
Tier before / after:	T4 / T4 (quantitative gate closure only).
No-overclaim statement:	Not a sharp practical error bar, $t=0$ H^2 rate, finite-oversampling exactness, solver certificate, PUBLISHED bundle, T5/T6/T7 promotion.
Next required action:	Integrate into the referee package and conduct the separate T5/T6 packaging and tier review.